

Smart delivery project

Main points:

Module A — Authentication & Users

Module B — Delivery Management

Module C — Driver Management

Module D — Dashboard & Frontend

Module E — AI Agent

Module A — Authentication & Users

Responsible for:

- Registration
- Login
- Logout
- Profiles
- Roles
- Permissions

Possible roles:

Admin

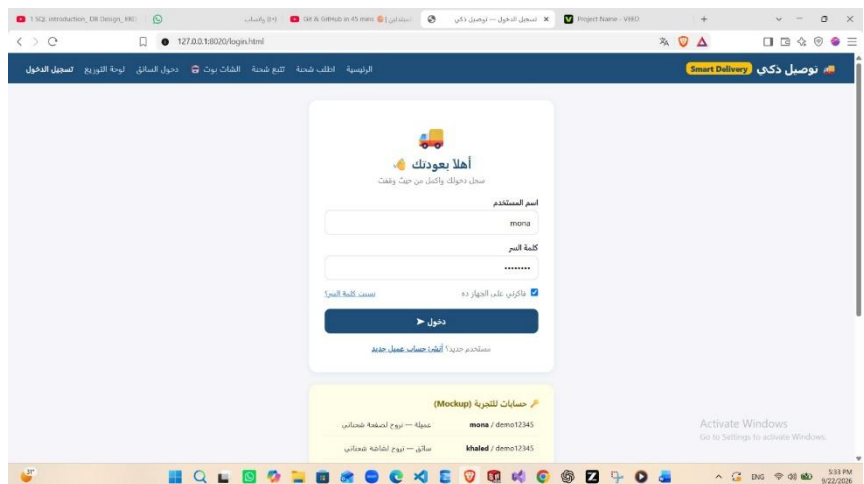
Manager

Driver

Customer

The specification requires

authentication and RBAC, with the AI inheriting the logged-in user's permissions.

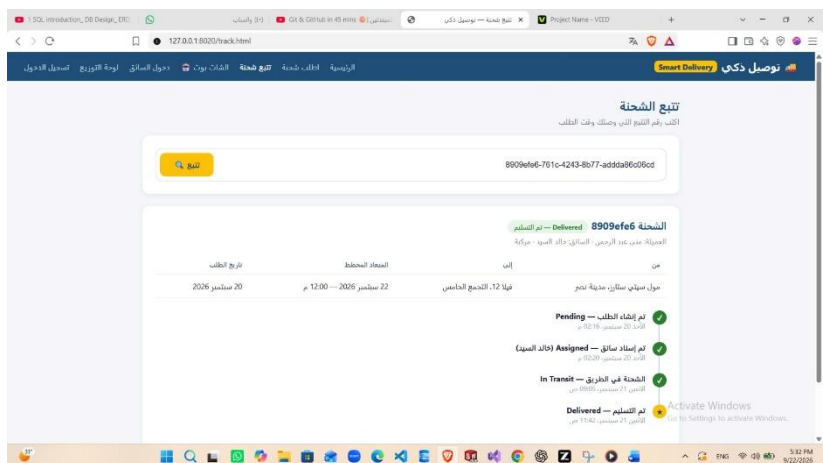


Module B — Delivery Management

This is the **core business module**.

It manages:

- Customers
- Shipments
- Delivery addresses
- Shipment status
- Delivery assignments
- Dates
- Priority
- Tracking



Module C — Driver Management

Responsible for:

- Driver profiles
- Driver availability
- Assigned deliveries
- Delivery history
- Updating shipment status

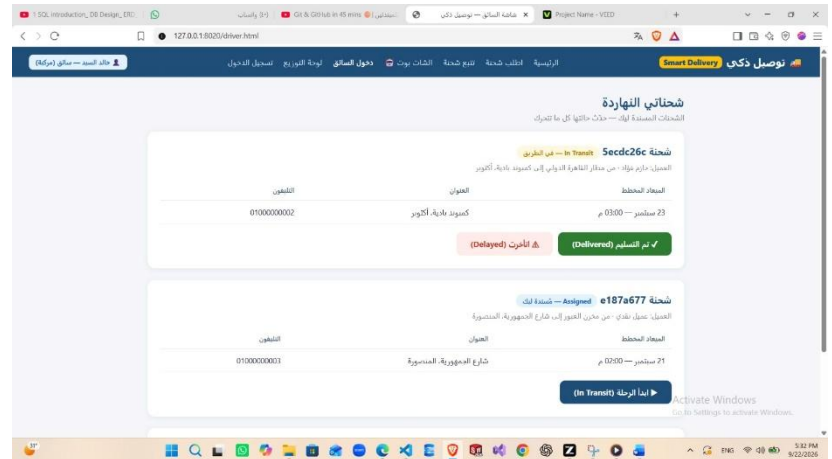
For example:

Driver: Mohamed

Status: Available

Assigned:

- #105 → In Transit
- #108 → Pending
- #111 → Picked Up



Module D — Dashboard & Frontend

This is where users interact with the system.

Different users should see different dashboards.

Manager:

Total Shipments: 250

Pending: 35

In Transit: 80

Delivered: 120

Delayed: 15

Driver:

My Deliveries

#105 Menouf In Transit

#108 Cairo Pending

#111 Shebin Delivered

Customer:

My Shipments

#105

Status: In Transit

Driver: Mohamed

Expected: Tomorrow

Module E — AI Agent

This is the most important special requirement.

The AI should have **tools**.

For example:

`get_delivery_status()`

`get_pending_deliveries()`

`get_delayed_deliveries()`

`get_available_drivers()`

`assign_delivery()`

`update_delivery_status()`

The AI could receive:

"Show me all delayed deliveries."

and choose:

`get_delayed_deliveries()`

Or:

"Reassign shipment 105 to an available driver."

The flow becomes:

User

↓

AI

↓

`assign_delivery(...)`

↓

Django validates user permission

↓

Django validates shipment + driver

↓

Database updated

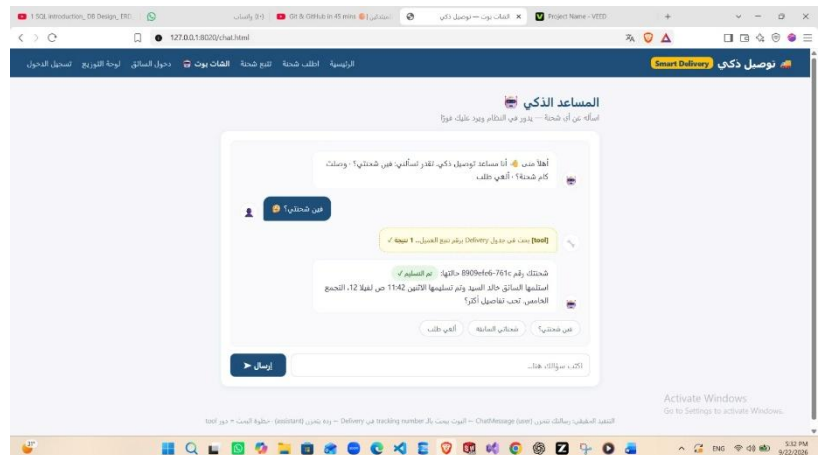
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Result → AI

↓

User

The specification specifically says AI tools need identity binding, permission checks, parameter validation, and confirmation for destructive actions.



The specification requires the system to manage deliveries, customers, drivers, delivery assignments, and shipment statuses, with the AI Agent able to analyze delayed/pending deliveries and perform actions such as reassignment or status updates through permission-checked Django tools.